

Electrodeposition of silicon nanotubes at room temperature using ionic liquid

Cite this: *Phys. Chem. Chem. Phys.*, 2013, **15**, 16446

Received 10th April 2013,
Accepted 7th August 2013

DOI: 10.1039/c3cp51522b

www.rsc.org/pccp

Jeremy Mallet,* Florie Martineau, Karine Namur and Michael Molinari

Electrodeposition inside an insulated nanoporous template using Room Temperature Ionic Liquids (RTILs) has recently been demonstrated as a promising alternative technique for synthesizing silicon nanowires due to low cost ambient growth conditions. An improvement of the method is shown here to produce Si nanotubes. A fine adjustment of electro-chemical parameters influencing ionic diffusion inside the nanopores of the template is demonstrated to preferably lead to the growth of Si nanotubes at the expense of Si nanowires. This study shows that electrodeposition in RTILs is a competitive process to grow high surface to volume ratio nanostructures at low cost and over a large scale. It also indicates a new prospect for the technique to grow and control nanostructures such as radial core-shell nanowires.

To decrease the size of existing technological devices, two types of nanostructures have been investigated in the past 15 years as building blocks of various nanoscale devices: carbon nanotubes and silicon nanowires (SiNWs). For specific future nanoscale devices,¹ such as sensors and batteries, the stable tubular shape of carbon nanotubes arising from the stable sp^2 hybridization of carbon makes them even more attractive because of their high surface-to-volume ratio. To increase the surface-to-volume ratio of silicon-based nanostructures, researchers have focused for the past 6 years on the possibility of growing silicon nanotubes (SiNTs), even though silicon primarily exhibits sp^3 hybridization, which preferably leads to the growth of SiNWs and not SiNTs.² Since the preliminary results published in 2002 by Sha *et al.*,³ who demonstrated the possibility of growing SiNTs through a vapor-liquid-solid process, many studies have demonstrated numerous perspectives and improvements of the silicon tubular shape for hydrogen storage,⁴ gas sensing,⁵ field emission⁶ and Li-ion batteries.⁷ In these studies, SiNWs or SiNTs were usually deposited using a vapor-liquid-solid process.^{1,3,8} This process can produce single-crystal silicon nanostructures with almost no impurities. However, the described silicon deposition methods

need to occur under a high vacuum and at a high temperature to reach the eutectic point for the metal catalyst, *i.e.*, silicon. Moreover, these processes are not easily transposable to future industrial large-scale syntheses.

In consequence, the development of an alternative low-cost method for growing SiNWs or SiNTs under room-temperature conditions that is easily transposable for building industrial large-scale integrated systems will be a challenge in the upcoming years. In our previous papers,⁹ we demonstrated the possibility of growing SiNWs through a simple and innovative technique: electrodeposition in a room-temperature ionic liquid. In this study, we describe an improvement to this method for the large-scale growth and control of the tubular shape of organized arrays of SiNTs using different growing parameters.

An interesting aspect of the electrodeposition technique is the possibility of growing nanomaterials with various geometries simply by changing the shape of the “nanoholes” in the insulated template; the only limitation to the possible geometries is that the electrolyte has to be able to wet the “nanoholes”.¹⁰ Among the various geometries, nanotubes are particularly attractive because they have a greater surface-to-volume ratio, can likely be used to observe quantum confinement with a small wall thickness and can be used to grow radial heterostructures by filling the nanotubes with another material. Consequently, the growth of SiNTs using electrodeposition in RTILs is highly attractive.

The possibility of growing metallic nanotubes through electrodeposition in an aqueous solution has been recently reported.¹¹ These recent results provide a way for obtaining this type of geometry using RTILs. Although the electrodeposition mechanism used for the growth of nanotubes in an aqueous solvent is still under debate, two major parameters appear to have a strong influence on the tubular growth: the conductive layer on the bottom side of the membrane (coverage and shape) and the gas evolution that occurs during the growth.¹¹ The tubular shape appears to be produced by the partial pore coverage provided by the conductive layer on the bottom side of the membrane and high gas evolution (mostly hydrogen through the reduction of water). Yoo *et al.*¹² explained that the tubular

UFR Sciences Exactes et Naturelles, Lab. de Recherche en Nanosciences (LRN),
51687 Reims, France. E-mail: jeremy.mallet@univ-reims.fr

shape can be stimulated if the conductive layer on the bottom side of the membrane penetrates the pores to form an initial tube structure, which would create a non-uniform electric field concentrated at the edge tip. Finally, Fukunaka *et al.*¹³ indicated that metallic nanotubes cannot be fabricated by electrodeposition if the growing grains readily become comparable to an employed pore size. Consequently, large pores with diameters of a few hundred nanometers are required. Based on these assumptions, Philippe *et al.*^{11c} demonstrated the possibility of controlling the geometry of metallic nanotubes in an aqueous solution using an insulating membrane with sufficiently large nanopores by varying the conductive layer and/or the electrodeposition parameters that affect the gas evolution and the electric field.

Taking into account these considerations obtained from studies performed in aqueous solution, SiNTs were successfully obtained through electrodeposition in an ionic liquid at room temperature (25 °C) using large nanopores of 400 nm. Fig. 1 displays an SEM image of the produced SiNTs after the dissolution of the PC membrane. The image shows a bunch of SiNTs grown in a 400 nm PC membrane with a high density.

The SiNTs are reasonably homogeneous in diameter (corresponding to the size of the nanoholes of the PC membrane). Depending on the deposition time, the length of the SiNTs is homogeneous and can be adjusted from a few hundred nanometers to a few microns using the same 400 nm PC membrane. To confirm and more clearly observe the tubular shape, top-view cross-sectional images of a few individual SiNTs were obtained. An example of these images (and the associated EDX line profile) is presented in Fig. 2. Only silicon and oxygen signals are detected with EDX. The silicon signal of the profile reveals that a sharp tubular shape with an average wall thickness of 120 nm was obtained under these deposition conditions. The wall of the nanotube is composed of pure silicon in the core with a thin layer of oxide on the edge in the limits of EDX detection. To determine the crystalline or amorphous nature of the 400 nm SiNTs electrodeposited in RTILs, Raman spectroscopy was performed on bunches of the

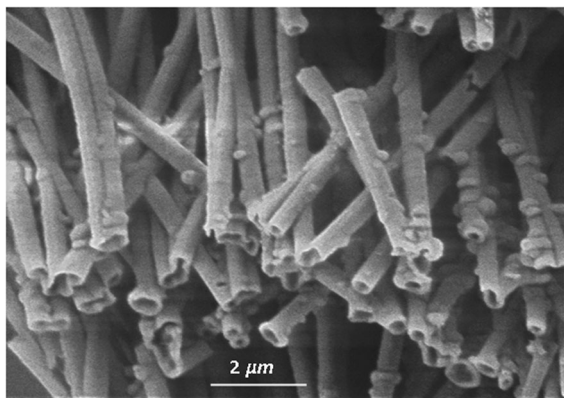


Fig. 1 SEM image of 400 nm SiNTs grown in 400 nm PC membranes for 2 hours at 25 °C. The small silicon bumps are caused by imperfections in the PC template and are always observed when this specific template is used.

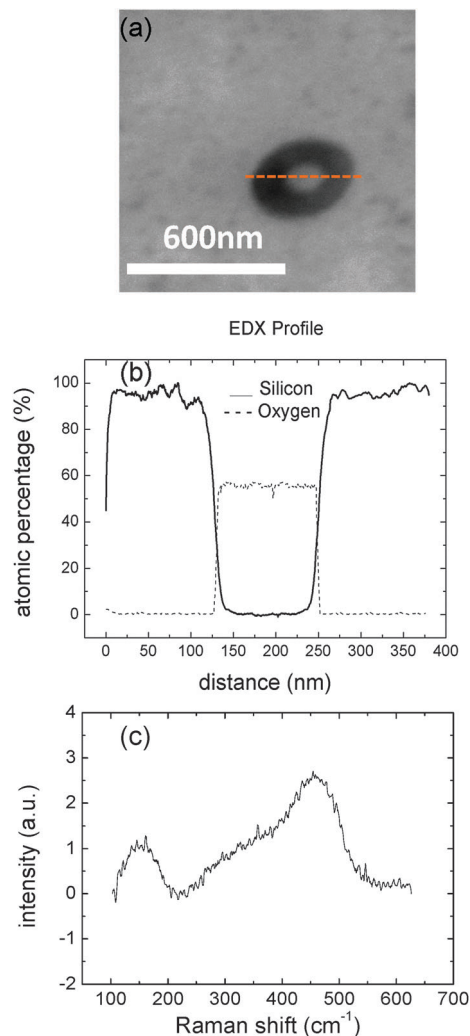


Fig. 2 Bright-field TEM top-view cross-sectional image of a 400 nm SiNT (a) and the related EDX silicon and oxygen profiles (b) along the dotted line. The image (a) and associated profile (b) confirm the tubular shape of the nanotube. Typical Raman spectrum (c) of SiNTs produced by electrodeposition in an ionic liquid at room temperature. The broad peak at 480 cm^{-1} is representative of amorphous silicon.

produced nanotubes. Fig. 2(c) shows a typical Raman spectrum with a broad peak at 480 cm^{-1} . This peak is representative of the presence of amorphous silicon. No peak at 520 cm^{-1} , characteristic of crystalline silicon, was detected whatever the sample. These results demonstrate the possibility of growing pure silicon nanotubes by electrodeposition in RTILs. To our knowledge, this is the first report of the synthesis of SiNTs at room temperature. To understand how the size and morphology of SiNTs produced by electrodeposition on RTILs can be controlled, the influence of parameters that are thought to control the tubular shape in an aqueous solution (gas evolution and conductive layer) was estimated. However, during the electrodeposition of silicon in P_{1,4} RTIL, no gas evolution was detected (mostly because the solvent has a significantly higher electrochemical window than water); hence, the formation of nanotubes cannot be explained by gas evolution. Consequently, we attempted to understand how the growth of the shape of

SiNTs can be controlled, as was previously described for metallic nanotubes in an aqueous solution. Thus, the conductive gold layer sputtered on the bottom side of the PC membrane was adjusted such that it completely or partially covered the pores to form an initial tube structure or adjusted to cover the pores by a flat surface.

One of the main differences between water and RTILs is the viscosity. Similarly to water, and despite the lack of reliable models to describe mass transport and material growth in RTILs, the viscosity of RTILs is thought to be strongly related to ionic conductivity and ionic diffusion.¹⁴ Besides, the conductivity of RTILs through nanopores of insulated polymer membranes is significantly modified due to interaction of these confined RTILs with charged surfaces. Electrostatic interactions of RTILs with nanopores seem to be the basis for the formation of ionic diodes rectifying transport of the constituent ions.¹⁵ Thus, the tubular shape obtained by electrodeposition in $P_{1,4}$ RTIL is certainly influenced by the ionic diffusion of silicon ions inside the nanopores resulting from the high viscosity of this RTIL relative to water (80-fold more viscous).

This assumption implies that the electrodeposition parameters that influence the diffusion of silicon ions inside the nanopores most likely influence the geometry of the nanotubes. To influence the diffusion regime or, more specifically, the concentration gradient of silicon ions between the bulk of the solution and the substrate interface, SiNTs were grown at various temperatures using different deposition regimes (steady-state current deposition or pulse deposition) and at various membrane thicknesses. Recently, Endres *et al.* reported the difficulty of growing thick $\text{Si}_x\text{Ge}_{1-x}$ films using the same $P_{1,4}$ RTIL because of chemical dissolution that resulted from the decomposition of the $P_{1,4}$ RTIL.¹⁶ This competition between electrochemical growth and chemical dissolution that was specifically found for $P_{1,4}$ RTIL for $\text{Si}_x\text{Ge}_{1-x}$ compound could also be an essential parameter for tubular growth.

Based on all of these considerations, SiNTs were elaborated by changing the electrodeposition parameters previously cited (*i.e.*, the Au layer and the diffusion regime). Fig. 3 displays two top-view images of 400 nm SiNTs grown under the same conditions (0.1 M SiCl_4 electrolyte at room temperature, applying a constant potential of -2700 mV *vs.* Pt) but different membrane thickness (the images were obtained after the dissolution of the PC membrane). Depending on the membrane thickness, the wall width can be varied between 40 nm (20 μm thick membrane) and 100 nm (5 μm thick membrane), on average. The inset of Fig. 3(b) displays, at a higher-magnification, the homogeneity of a 100 nm wall width of a SiNT. These preliminary results demonstrate the possibility of electrodepositing SiNTs in RTILs, and shows that the membrane thickness can be varied to modify the wall thickness of the tubes.

Besides, it has already been demonstrated in previous studies that the membrane thickness plays an important role in the diffusion regime during nanoporous template electrodeposition in aqueous solution.¹⁷ Considering that these studies can be transposed to an electrodeposition mechanism in ionic liquids, it would confirm that the diffusion regime has

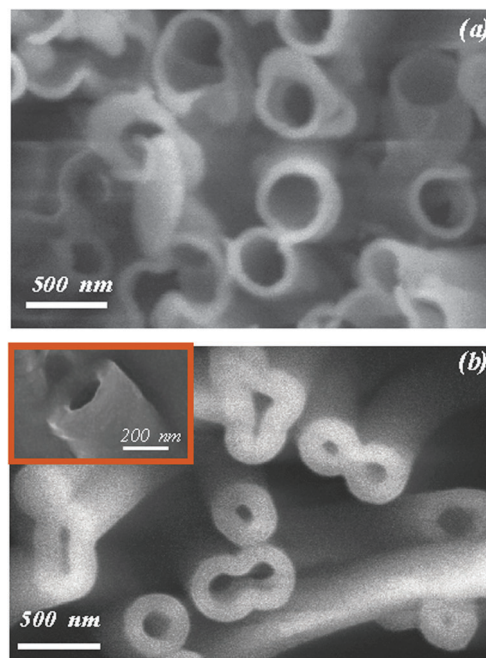


Fig. 3 Images of SiNTs electrodeposited in 400 nm PC membranes at room temperature applying a constant potential of -2.7 V *vs.* Pt. (a) 40 nm wall thickness (20 μm thick template) and (b) 100 nm wall thickness (5 μm thick template).

an impact on the geometry of the nanotubes. A lower diffusion of the ions (a thick membrane for instance) appears to decrease the wall thickness.

However, the gold layer does not exhibit any strong influence. This finding represents a major difference between the method described in this study and the synthesis of metallic nanotubes in an aqueous solution; it is still possible to obtain complete SiNTs using a flat surface on the bottom, whereas it is not possible to obtain this shape in an aqueous solution under the same conditions.^{11b}

Although the growth mechanism of SiNTs in $P_{1,4}$ RTIL is still unclear, it appears that these nanotubes are formed by competition between the growth rate and the ionic diffusion inside the nanopores of the insulating template (which is quite low for two main reasons: the high viscosity of the $P_{1,4}$ ionic liquid and the diffusion regime inside the nanopores). If membrane thicknesses enable us to play with the diffusion regime, the nature of the ionic liquid and the temperature of the experiments could also be used to change the characteristics of the NTs.

Future investigation is necessary to further elucidate and control this growth mechanism of SiNTs and to eventually demonstrate the feasibility of producing other nanostructures, such as the growth of radial SiGe nanowires.

Conclusions

In conclusion, the growth of Si nanostructures *via* electrodeposition in a room-temperature ionic liquid has been improved. Large-scale arrays of SiNTs were obtained using this method.

Moreover, the preliminary results show that the length and thickness of the SiNT geometry can be controlled by varying the electrodeposition parameters.

Experimental

The SiNTs were prepared in a three electrode Teflon cell. Commercially available PolyCarbonate (PC) membranes with perpendicularly pierced pores with a diameter of 400 nm served as the template for the deposition of the nanotubes (membranes are provided by it4ip s.a.). Different membrane thicknesses were used ranging from 5 μm to 20 μm for the same porosity (10^8 pores per cm^2). Counter electrode and reference electrode are respectively a Pt wire and a Pt grid heated in a hydrogen flame prior to use in order to minimize possible surface contamination. The potential of the Pt quasi reference electrode was sufficiently stable for the experiment done in the study. The electrolytic bath was composed of 1 butyl 1 methylpyrrolidinium bis(trifluoromethylsulfonyl)imide ($\text{P}_{1,4}$; Solvionic S.A.) and SiCl_4 (Aldrich) (1 M). Because the solution is very sensitive to water and oxygen, it was prepared inside a glovebox (Inert Lab Glovebox, Innovative Technology, Inc.) containing less than 1 ppm of water and less than 1 ppm of oxygen. The electrodeposition was also performed inside the glovebox. The SiNTs have been electrodeposited with a radiometer analytical potentiostat (PGZ 301).

Acknowledgements

This research was supported by the SYNAPSE and by the Nano'Mat programs funded by la Region Champagne Ardenne, FEDER, and the DRRT Champagne Ardenne. The authors thank the company IT4IP S.A. (Belgium) for providing the polycarbonate membranes.

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